MOLECULAR MECHANISMS OF EXCITATORY GLYCINE RECEPTORS

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ABSTRACT

Within the ionotropic glutamate receptors **iGluR** family, **GluN1/GluN3A** forms an atypical subtype of **NMDA receptors** (NMDARs). Unlike conventional NMDARs, both the GluN3A and GluN1 subunits are insensitive to glutamate. Together they assemble as excitatory glycine-gated receptors (**eGlyR**), where both activation and desensitization are driven solely by glycine, resulting in highly distinctive functional properties[1,2]. This unique activation mechanism results in **rapid desensitization**[2] of receptor-mediated currents, although the underlying molecular mechanism remains poorly understood despite recent advances. Recent structural studies have revealed a **susbtantial rearrangement** of the ligand binding domain (**LBD**) layer upon glycine binding[3,4], resembling the desensitized state of kainate receptors (KAR)[5]. It has also been observed that the N-Terminal domain (**NTD**) of GluN3A is **highly flexible**[4], suggesting a potential role in modulating these conformational changes. Nevertheless, the precise state transitions as well as the **molecular determinants** of the receptor dynamics remain unclear.
To further explore the receptor function, we aim to characterize the different **conformational states** of the receptor as well as to identify the **structural determinants** governing activation and desensitization of eGlyR by integrating *in-silico*, biomolecular, mutagenesis, pharmacological and electrophysiological approaches.
Overall, our work seeks to provide new insights into the **mechanistic basis** of GluN1/GluN3A receptor function. Given that eGlyRs are thought to regulate neuronal excitability in various brain regions[6,7] and have been implicated in disorders such as schizophrenia, bipolar disorder, and epilepsy[6,8], elucidating their mechanism could contribute significantly to the development of novel pharmacological strategies and therapeutic interventions.

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